Lecture Notes for Math 243: Calculus III

Last updated: September 4, 2026

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1 2026-08-24 | Week 01 | Lecture 01

This lecture is based on textbook section 6.3. "Arc length"

> ***The nexus question of this lecture:*** *How do we compute the length of a curve, assuming the curve is the graph of a function?*

1.1 Arc Length

First, recall the definition of ***derivative***:

$$f'(x) := \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

or equivalently,

$$f'(x) = \lim_{\tilde{x} \to x} \frac{f(\tilde{x}) - f(x)}{\tilde{x} - x}$$

In other words, the derivative is obtained by computing

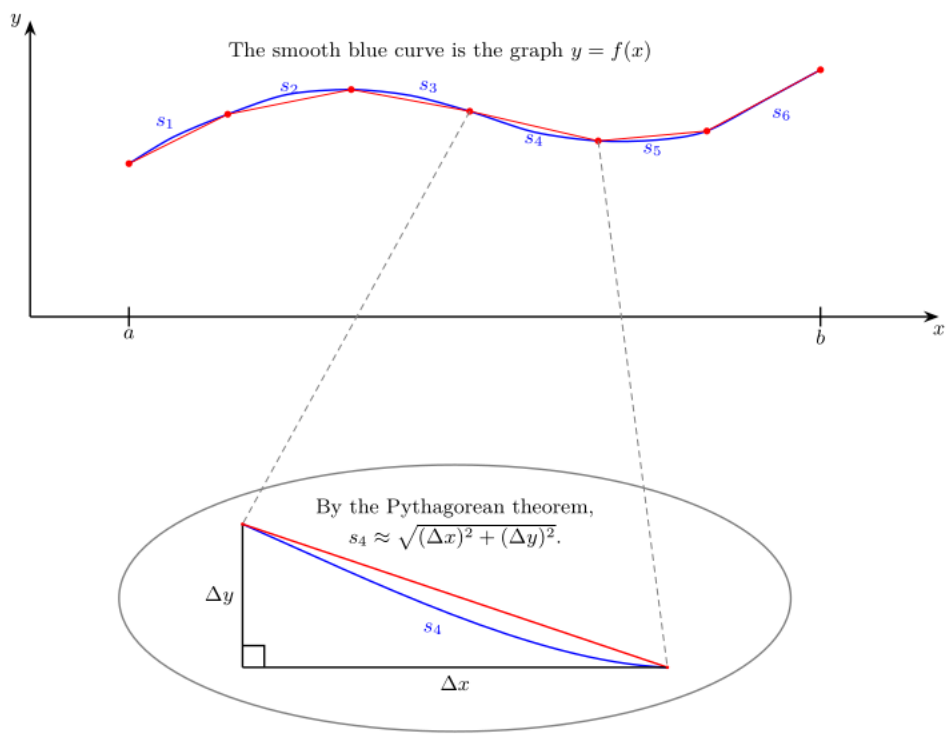
$$\frac{\text{change in } y\text{-value}}{\text{change in } x\text{-value}} = \frac{\Delta y}{\Delta x}$$

as $\Delta x \to 0$.

One of the central ideas in calculus is to approximate curvy things with flat things. For example, if faced with a curve, approximate it with straight lines. For example,

Problem: I am an ant, and I want to travel from a to b by walking on the graph of $f(x)$. How far must I walk?

Strategy: Cheat a little: think of the ant walking in a bunch of straight lines that are pretty close to the curve.



In this example, the length of the curve $y = f(x)$ from $x = a$ to $x = b$ can be decomposed as

$$(\text{Arc Length}) = s_1 + s_2 + \ldots + s_6,$$

where

$$s_i = (\text{length of segment } i)$$

Each segment length can be approximated by a straight line. For example,

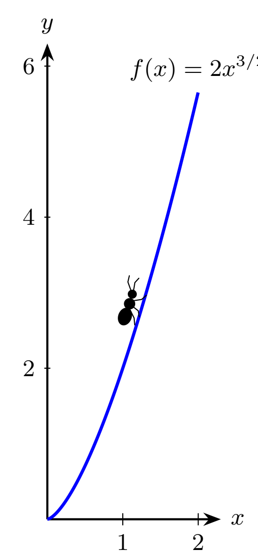
$$\begin{aligned}
s_4 &\approx \sqrt{(\Delta x)^2 + (\Delta y)^2} && \text{since the line segment length approximates the curve length} \\
&= \sqrt{\left[1 + \left(\frac{\Delta y}{\Delta x}\right)^2\right] (\Delta x)^2} \\
&= \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2}\, \Delta x \\
&\approx \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx && \text{this holds when } \Delta x \approx 0\text{, so long as the curve is smooth} \\
&= \sqrt{1 + [f'(x)]^2} dx && \text{by change of notation}
\end{aligned}$$

This example motivates the following theorem.

Theorem 1 (Arc length formula)**.** *Let f be a differentiable function on the interval $[a, b]$, and assume that f' is continuous on $[a, b]$. Then the arc length of f from $x = a$ to $x = b$ is*

$$(\textit{Arc Length}) = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

Example 2 (Arc length)**.** Find the length of the curve defined by $f(x) = 2x^{3/2}$, where x ranges from $x = 0$ to $x = 2$.



Solution: $f'(x) = 3x^{\frac{1}{2}}$, so

$$\begin{aligned}
(\text{Arc length}) &= \int_0^2 \sqrt{1 + [f'(x)]^2} dx \\
&= \int_0^2 \sqrt{1 + 9x} dx
\end{aligned}$$

which we can solve with a u-substitution, taking $u = 1 + 9x$. Doing this gives a length of about 6.06.

End of Example 2. $\square$

The ***hyperbolic cosine*** and ***hyperbolic sine*** functions are defined as

$$\cosh(x) := \frac{e^x + e^{-x}}{2}, \qquad \sinh(x) := \frac{e^x - e^{-x}}{2}.$$

These two functions are "nice" in that they behave in some respects similarly to the standard sine and cosine functions; in particular, cosh is an even function (like $\cos(x)$) and $\sinh(x)$ is odd (like $\sin(x)$).

One can check they satisfy:

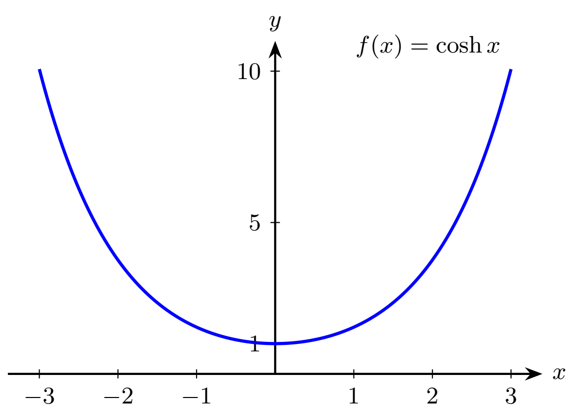
$$\cosh'(x) = \sinh(x), \qquad \sinh'(x) = \cosh(x). \tag{1}$$

Also, we have the following identity:

$$\cosh^2(x) - \sinh^2(x) = 1. \tag{2}$$

A hanging chain forms a curve known as a ***catenary***. The general formula of a catenary is a function involving hyperbolic cosine; in the next example, we'll consider the simplest case.

Example 3 (Arc length)**.** Suppose $f(x) = \cosh(x)$ represents a chain from two endpoints at $x = -3$ and $x = 3$. Find the length of the chain.



$$\begin{aligned}
\text{(length of chain)} &= \int_{-3}^{3} \sqrt{1 + \left[\cosh'(x)\right]^2} dx \\
&= \int_{-3}^{3} \sqrt{1 + \sinh^2(x)} dx && \text{since } \cosh'(x) = \sinh(x) \text{ by Eq. (1)} \\
&= \int_{-3}^{3} \sqrt{\cosh^2(x)} dx && \text{by Eq. (2)} \\
&= \int_{-3}^{3} \cosh(x) dx && \text{since } \cosh(x) > 0 \text{ for all } x \in \mathbb{R} \\
&= [\sinh(x)]_{x=-3}^{x=3} && \text{since } \sinh'(x) = \cosh(x) \text{ Eq. (1)} \\
&= \sinh(3) - \sinh(3) \\
&\approx 20.036
\end{aligned}$$

End of Example 3. $\square$

Tips

- Check that your answer is positive. Arc length is never negative.
- These problems are usually contrived so that you can actually do the integral. If the integral looks took hard, double check your work. Also, there might be a trick to the integral.

2 2026-08-26 | Week 01 | Lecture 02

This lecture is based on textbook section 6.3 "Arc Length", and section 11.1 "Parametric Equations and Polar Coordinates". Please read 6.3 and begin start reading 11.1.

Example 4 (Warm up problem: Chain rule practice)**.** Find the derviative of

$$f(x) = \log\left(\frac{x^2}{\sin x}\right).$$

Solution:

$$\begin{aligned}
f'(x) &= \frac{d}{dx}\left[\log\left(\frac{x^2}{\sin x}\right)\right] \\
&= \frac{\sin x}{x^2}\frac{d}{dx}\left[\frac{x^2}{\sin x}\right] && \text{by the chain rule} \\
&= \frac{\sin x}{x^2}\left(\frac{2x\sin x - x^2\cos x}{(\sin x)^2}\right) && \text{by the quotient rule} \\
&= \frac{2}{x} - \cot x && \text{after simplification.}
\end{aligned}$$

End of Example 4. □

2.1 Arc length - examples

Example 5 (Arc length)**. Problem:** Let

$$f(x) = \frac{x^3}{3} + \frac{1}{4x} + 7,$$

for $1 \le x \le 2$. earlier examples.

Solution: We want to apply Theorem 1. To do this, we first compute $1 + [f'(x)]^2$:

$$\begin{aligned}
1 + [f'(x)]^2 &= 1 + \left(x^2 - \frac{1}{4}x^{-2}\right)^2 \\
&= x^4 + \frac{1}{2} + \frac{1}{16}x^{-4} \\
&= \left(x^2 + \frac{1}{4}x^{-2}\right)^2.
\end{aligned}$$

The last step, where we write the equation as a perfect square, is kinda hard to see. But since we're familiar with Example 3, we know to possibly expect something like that.

Now, we can apply the formula from Theorem 1:

$$\begin{aligned}
\text{Arc Length} &= \int_1^2 \sqrt{1+[f'(x)]^2}dx \\
&= \int_1^2 \sqrt{\left(x^2+\frac{1}{4}x^{-2}\right)^2}dx \\
&= \int_1^2 x^2+\frac{1}{4}x^{-2}dx \\
&= \left[\frac{x^3}{3}-\frac{1}{4}x^{-1}\right]_1^2 \\
&= \left(\frac{8}{3}-\frac{1}{8}\right)-\left(\frac{1}{3}-\frac{1}{4}\right) \\
&\approx 2.46.
\end{aligned}$$

End of Example 5. ☐

Example 6. Find the arc length of

$$f(x)=\frac{x^3}{12}+x^{-1}, \qquad 1\le x\le 4$$

Solution: For the complete solution, see Example 2 in section 6.3 of the textbook. We sketch the approach here.

The key step is to factor

$$\begin{aligned}
1+[f'(x)]^{-2} &= \frac{x^4}{16}+\frac{1}{2}+x^{-4} \\
&= \left(\frac{x^2}{4}+x^{-2}\right)^2
\end{aligned}$$

The arc length integral then reduces to

$$\int_1^4 \left(\frac{x^2}{4}+x^{-2}\right)dx$$

which evaluates to 6.

End of Example 6. ☐

2.2 Introduction to parametric equations

This section is based on chapter 11.1 in the textbook.

I am an ant walking around the whiteboard, and I'm covered in ink. My path is described in the following way:

- The variable t measures time. We start at $t=0$ and end at time $t=T_{\text{end}}>0$.
- At each moment in time t, the number $x(t)$ represents my x-coordinate
- And the number $y(t)$ represents my y-coordinate.

The ***parametric equation*** is the picture I draw by walking around the whiteboard.

Definition 7 (Parametric Equations and Curves)**.** Suppose x and y are given as functions

$$x=f(t), \qquad y=g(t)$$

over an interval of t-values I. Then as t varies over I, the set of all points of the form

$$(x, y) = (f(t), g(t)), \tag{3}$$

as t varies over I, is called a ***parametric curve***. The equations in Eq. (3) are called the ***parametric equations*** of the curve.

The variable t is called the ***parameter***. If $I = [a, b]$, then the ***initial point*** is $(f(a), g(a))$ and the ***terminal point*** is $(f(b), g(b))$.

Notation 8. In "set-builder notation," the graph of the parametric equations $x = f(t)$ and $y = g(t)$ is the set

$$\{(f(t), g(t)) : t \in I\}.$$

For those unfamiliar with this notnation: the colon ':' is read as "such that" and 'ϵ' is an abbreviation of "is an element of"

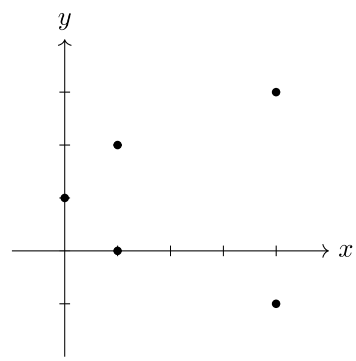
Example 9 (Plotting parametric functions)**.** Plot the graph of the parametric equations

$$\begin{cases} x = t^2 \\ y = t + 1 \end{cases}$$

for t in $I = [-2, 2]$.

Solution: Start by plotting some points by making a table.

t	$x = t^2$	$y = t + 1$
-2	4	-1
-1	1	0
0	0	1
1	1	2
2	4	3



Once we get a sense of what the curve looks like, we can draw a curve through the points.

- If we replace $I = [-2, 2]$ by $I = \mathbb{R}$, then this corresponding graph would trace out a sideways opening parabola defined by the equation $x = (y - 1)^2$.

Question: can we write y a function of x?
Answer: No, because the graph doesn't satisfy the vertical line test.

Question: can we express this as an equation involving only x and y?
Answer: yes, we the parabola is $x = (y - 1)^2$, so the parametric curve is can be written as

$$x = (y - 1)^2, \qquad 0 \leq x \leq 4.$$

End of Example 9. $\square$

3 2026-08-28 | Week 01 | Lecture 03

This lecture is based on section 11.1 in the textbook

Reading assignment: Section 11.1 in the text book.

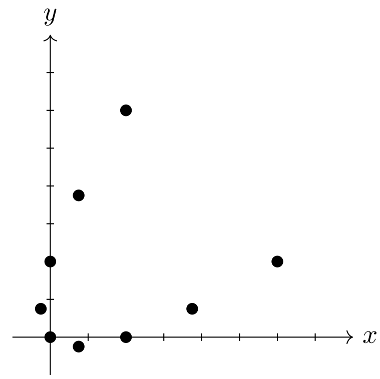
3.1 Plotting parametric equations

Example 10. *Question:* Plot the graph of the parametric equation

$$r(t) = (t^2 + t, t^2 - t), \quad -2 \le t \le 2$$

Solution: Again, we plot some points by making a table.

t	$x = t^2 + t$	$y = t^2 - t$
-2	2	6
$-\frac{3}{2}$	.75	3.75
-1	0	2
$-\frac{1}{2}$	$-\frac{1}{4}$	$\frac{3}{4}$
0	0	0
$\frac{1}{2}$	$\frac{3}{4}$	$-\frac{1}{4}$
1	2	0
$\frac{3}{2}$	3.75	$\frac{3}{4}$
2	6	2



The graph of the parametric equations is obtained by drawing the curve through the plotted points.

End of Example 10. □

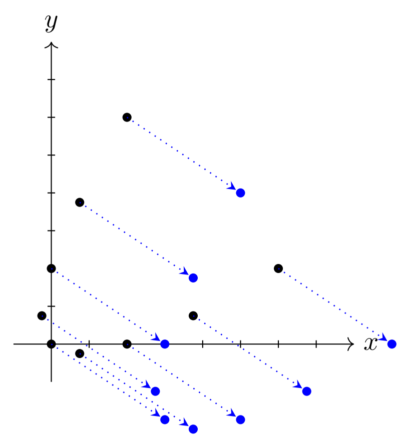
Example 11. Find the parametric equations that shift the graph from Example 10 to the right by 3 and down by 2.

Solution: The graphs of parametric equations are easy to shift. The parameteric equations defining the shifted graph are

$$r(t) = (t^2 + t + 3, t^2 - t - 2)$$

We've simply added 3 to the x-value and subtracted 2 from the y-value. To see why this is the case, observe what happens to the points in the plot from Example 10:

t	$x = t^2 + t + 3$	$y = t^2 - t - 2$
-2	$2+3$	$6-2$
$-\frac{3}{2}$	$.75+3$	$3.75-2$
-1	$0+3$	$2-2$
$-\frac{1}{2}$	$-\frac{1}{4}+3$	$\frac{3}{4}-2$
0	$0+3$	$0-2$
$\frac{1}{2}$	$\frac{3}{4}+3$	$-\frac{1}{4}-2$
1	$2+3$	$0-2$
$\frac{3}{2}$	$3.75+3$	$\frac{3}{4}-2$
2	$6+3$	$2-2$



Here we have shifted every point to the right by 3 and down by 2, as shown by the arrows. Drawing a curve through the blue points now gives us the curve of the new, shifted, parametric equations. (Not shown in figure).

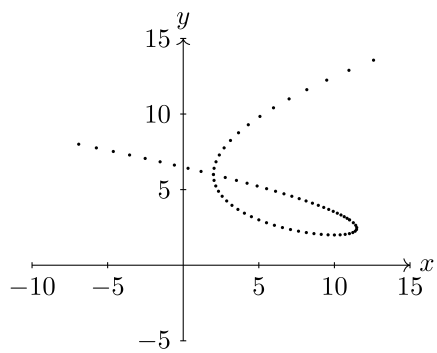
End of Example 11. □

Parametric functions have the advantage that we can use them to plot curves which cannot be plotted by "$y = f(x)$" type graphs.

Example 12. Here's an example of a plot of the parametric curve

$$r(t) = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} t^3 - 5t^2 + 3t + 11 \\ t^2 - 2t + 3 \end{pmatrix}$$

This one would be tedious to plot by hand, so instead I plotted a bunch of points with a computer, with t varying on the interval $I = [-1.5, 4.5]$.



The parametric equation can be thought of as tracing the trajectory of a particle as it moves over time. In this example, t represents time. At time $t = -1.5$, the particle is at point $(x, y) = (-8.125, 8.25)$, and starts heading southeast. It then turns around and loops over its previous path, before heading northeast.

It would be impossible to represent this trajectory using a function $y = f(x)$, becausethe curve fails the vertical line test. This is one advantage of parametric curves.

The particle has the same position at times $t = -1$ and $t = 3$. At these times, the particle is at the point

$$(x, y) = (2, 6).$$

Thus, it is at this point that the curve intersects itself, a phenomenon known as a ***singularity***.

End of Example 12. □

3.2 Converting between parametric equations and Cartesian equations

In contrast to parametric equations, an equation is said to be in ***rectangular form*** (aka: ***Cartesian form***) if it takes the form $y = f(x)$ (or, less commonly, $x = f(y)$) for some function f. We've just seen in Example 12 that parametric equations offer significantly more flexibility when graphing.

In fact, it is always possible to convert an equation from rectangular form into parametric form. We illustrate this with the next example:

Example 13 (Converting from rectangular form to parametric form)**.** Consider the equation $y = x^2$, which is in rectangular form. This is an upward-opening parabola. There are many ways to represent this in parametric form. The simplest is with the parametric equations

$$r(t) = (t, t^2), \qquad \infty < t < \infty.$$

End of Example 13. □

Example 14 (Converting from parametric form to rectangular form)**.** Sometimes (but not always) we can also convert from parametric to rectangular form. Consider the parametric equations

$$\begin{cases} x = t^2 \\ y = 7 + 5t \end{cases}$$

This is done by "eliminating" the parameter t, in the following way:

Since $y = 7 + 5t$, it follows that

$$t = \frac{y-7}{5}.$$

Plugging this into $x = t^2$ gives

$$\begin{aligned} x &= \left(\frac{y-7}{5}\right)^2 \\ &= \frac{1}{25}\left(y^2 - 14y + 49\right) \\ &= \frac{1}{25}y^2 - \frac{14}{25}y + \frac{49}{25}. \end{aligned}$$

This equation has the form $x = f(y)$, and therefore is in rectangular (i.e., Cartesian) form.

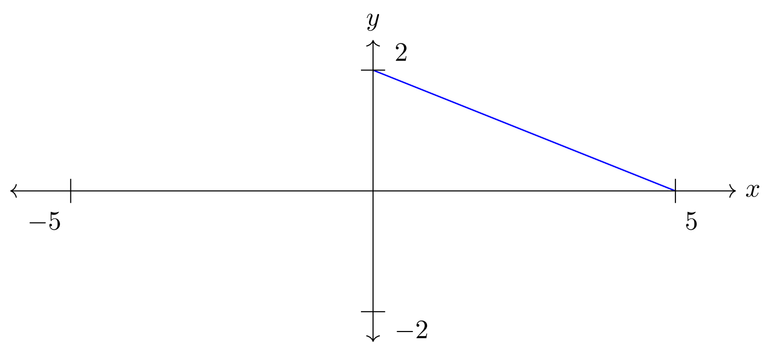
End of Example 14. $\square$

Here's another example

Example 15 (Converting to Cartesian form: eliminating a parameter)**.** Consider the parametric equations

$$\begin{cases} x = 5\cos^2(\theta) \\ y = 2\sin^2(\theta) \end{cases}, \qquad 0 \leq \theta \leq 2\pi.$$

I plotted this using a computer and got the following. It is a line segment:



Question: Convert these parametric equations into Cartesian form, ie a function of the form $y = f(x)$.

Solution: Before we start, recall the identity $\sin^2(\theta) + \cos^2(\theta) = 1$ for all θ. using this, we have:

$$\begin{aligned} y &= 2\sin^2(\theta) \\ &= 2\left(1 - \cos^2(\theta)\right) \end{aligned}$$

Now, $x = 5\cos^2(\theta)$, so $\cos^2(\theta) = \frac{x}{5}$. Plugging this into the above equation gives:

$$y = 2\left(1 - \frac{x}{5}\right).$$

This is an equation of a line.

But we are not done! We are trying to plot only a line *segment*, not the full line. So we need to specify that we are restricting the line to only those x in the interval $[0, 5]$. So our solution (in Cartesian form) is

$$\begin{cases} y = 2\left(1 - \frac{x}{5}\right) \\ 0 \leq x \leq 5 \end{cases}$$

End of Example 15. $\square$

4 2026-08-31 | Week 02 | Lecture 04

Reading assignment: Chapter 11.2 "Calculus in Parametric Curves"

This is probably the most important lecture I will give this semester, so please try to understand it in detail.

4.1 Ellipses

Definition 16 (Ellipse)**.** An ***ellipse*** centered at (h,k) with horizontal radius $a > 0$ and vertical radius $b > 0$ has equation

$$\left(\frac{x-h}{a}\right)^2 + \left(\frac{y-k}{b}\right)^2 = 1.$$

Example 17 (Eliminating a parameter)**.** Eliminate the parameter t in the parametric system of equations

$$\begin{cases} x = 4\cos(t) + 3 \\ y = 2\sin(t) + 1 \end{cases} \tag{4}$$

The trig identity $\sin^2(t) + \cos^2(t) = 1$ served us well in Example 15, so let's try to use that. First, solve for $\sin(t)$ and $\cos(t)$:

$$\sin(t) = \frac{y-1}{2}, \qquad \cos(t) = \frac{x-3}{4}. \tag{5}$$

By the trig identity, we have

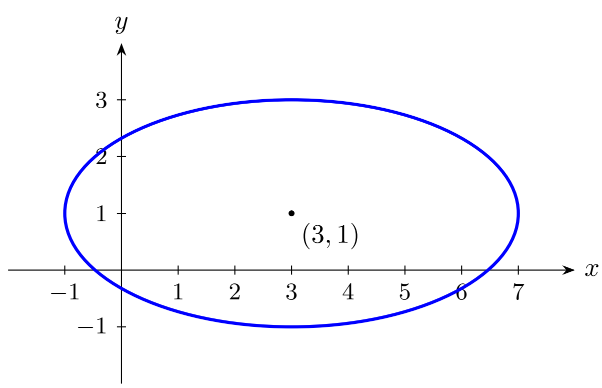
$$\cos^2(t) + \sin^2(t) = 1.$$

Substituting in the formulas from Eq. (5) gives:

$$\left(\frac{x-3}{4}\right)^2 + \left(\frac{y-1}{2}\right)^2 = 1,$$

This is the equation of an ellipse, with center $(3,1)$ and with horizontal and vertical axes 4 and 2 respectively. These numbers can be read off of Eq. (4).

The plot is



At time $t = 0$, the ant starts at position $(x, y) = (7, 1)$ and then marches counter-clockwise around the ellipse. At $t = \pi/2$, it reaches the point $(3,3)$. At time $t = \pi$ it reaches $(-1,1)$. At time $t = 2\pi$, the ant returns to the point where it started $(7,1)$.

End of Example 17. $\square$

4.2 Parameterized curves: velocity, speed, and direction

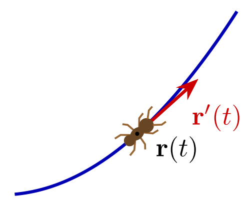
Suppose an ant is walking along the curve in the xy-plane defined by the parametric equation

$$\mathbf{r}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}, \qquad a \le t \le b$$

where t represents time.

Three questions:

- what is the ant's velocity?
- how fast is the ant walking?
- what direction is the ant facing?



Answers:

$$\text{velocity} = \mathbf{r}'(t), \qquad \text{speed} = |\mathbf{r}'(t)|, \qquad \text{direction} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}.$$

We'll explain them one-by-one.

Velocity. First, recall that ***velocity*** is defined as the derivative of position with respect to time. The ant's position at time t is $\mathbf{r}(t)$. Therefore its velocity at time t is

$$\mathbf{v}(t) = \mathbf{r}'(t) = \begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix}. \tag{6}$$

To see geometrically why this is natural, let $\Delta t > 0$ represent a small amount of time. The ant's position at time t and $t + \Delta t$ are

$$\mathbf{r}(t) \quad \text{and} \quad \mathbf{r}(t + \Delta t).$$

The ant's average velocity between times t and $t + \Delta t$ is

$$(\text{average velocity}) = \frac{\text{displacement}}{\text{length of time}} = \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}.$$

As $\Delta t \to 0$, the average velocity converges to the instantaneous velocity $\mathbf{v}(t)$; the right-hand side converges to $r'(t)$ (by the definition of derivative). Hence

$$\mathbf{r}'(t) = (\text{velocity at time } t).$$

The vector $\mathbf{r}'(t)$ contains two pieces of information: how fast the ant is moving and which way it is moving.

Speed. Recall that the ***magnitude*** of a vector $\mathbf{v}$, denoted $|\mathbf{v}|$ is the length of the vector. That is, if $\mathbf{v} = (v_1, \ldots, v_n)$, then

$$|\mathbf{v}|= \sqrt{v_1^2 + \ldots + v_n^2}.$$

The ***speed*** of a moving object is the magnitude of its velocity. Therefore, the ant's speed at time t is

$$|\mathbf{v}(t)| = |\mathbf{r}'(t)| = \sqrt{x'(t)^2 + y'(t)^2}.$$

Direction. We know the velocity points in the direction the ant is traveling. The vector

$$\frac{\mathbf{v}(t)}{|\mathbf{v}(t)|} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

is simply velocity divided by speed (provided the speed is nonzero!). This vector points in the same direction as the velocity and has length 1, so it records the direction of motion without recording the speed. (This is useful for certain applications.)

4.3 Arc length of a parameterized curve

If the ant walks at a constant speed, then

$$\text{(distance traveled)} = \text{(speed)} \times \text{(time)}.$$

If its speed varies, break the journey into many tiny pieces. During each tiny time interval $[t, t+\Delta t]$, her speed is nearly constant, so

$$\text{(distance traveled from time } t \text{ to } t+\Delta t) \approx \underbrace{|\mathbf{r}'(t)|}_{\text{speed at } t} \times \Delta t.$$

Formally, if the ant starts at time $t = a$ and ends at time $t = b$, we break the time interval into n pieces:

$$a = t_0 < t_1 < t_2 < \cdots < t_n = b$$

and put

$$\Delta t_i = t_{i+1} - t_i.$$

Then

$$\text{(Approx. distance traveled)} = \sum_{i=0}^{n-1} |\mathbf{r}'(t_i)| \Delta t_i$$

Geometrically, each term $|\mathbf{r}'(t_i)|\Delta t_i$ is the distance the ant would travel if it moved in a straight line, at constant speed $|\mathbf{r}'(t_i)|$, for Δt_i units of time. Stringing these segments together gives a path made of straight pieces which approximates the curve, in the same spirit as the polygonal approximation in Section 1.1.

As $n \to \infty$ and the time intervals get smaller (i.e., as $\max_i \Delta t_i \to 0$), the approximation gets better and better, and the right hand side (which is a Riemann sum) converges to an integral:

$$\text{(Distance traveled)} = \int_a^b |\mathbf{r}'(t)| dt.$$

Since $|\mathbf{r}'(t)| = \sqrt{[f'(t)]^2 + [g'(t)]^2}$, we obtain the following theorem:

Theorem 18 (Arc Length of a Parameterized Curve)**.** *Consider the parameterized curve*

$$\mathbf{r}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}, \qquad a \le t \le b.$$

Then the distance traveled along the curve is

$$\text{(Distance)} = \int_a^b |\mathbf{r}'(t)| dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2}\, dt, \tag{7}$$

provided that the curve is smooth.

Further, if the parameterization traces the curve only once, this is the arc length of the curve.

Remark 19. *The integrals in Eq.* (7) *are typically difficult to compute by hand. The meaning of smooth will be discussed later.*

End of Remark 19. □

5 2026-09-02 | Week 02 | Lecture 05

Reading assignmnt: Section 11.2 in the textbook.

5.1 Arc length and Distance Examples

Example 20 (Arc length of a circle)**.** Consider the circle parameterized by the equationss

$$\mathbf{r}(t) = \begin{bmatrix} 3\cos(t) \\ 3\sin(t) \end{bmatrix}, \qquad 0 \le t \le 2\pi.$$

What is the arc length?

Solution: The velocity of an ant traversing the curve is

$$\mathbf{r}'(t) = \begin{bmatrix} -3\sin(t) \\ 3\cos(t) \end{bmatrix}.$$

Therefore the speed is

$$|\mathbf{r}'(t)| = \sqrt{(-3\sin t)^2 + (3\cos t)^2} = \sqrt{9[\sin^2(t) + \cos^2(t)]} = 3,$$

where we have used the identity $\sin^2(t) + \cos^2(t) = 1$.

By Theorem 18, integrating the speed over time gives distance traveled:

$$\begin{aligned} (\text{distance}) &= \int_0^{2\pi} |\mathbf{r}'(t)|\, dt \\ &= \int_0^{2\pi} 3\, dt \\ &= 6\pi. \end{aligned}$$

Since the ant walks around the circle only once in the time interval $[0, 2\pi]$, the distance it traveled is the arc length.

Taking a step back. What was this question really asking? It was asking for the circumference of a circle with radius 3 (since that's the curve traced out by the equations). Using the formula

$$\text{circumference} = 2\pi(\text{radius}),$$

our answer 6π makes sense.

End of Example 20. □

Example 21 (Arc length)**.** What is the arc length of the curve parameterized by

$$\mathbf{r}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} \frac{7}{3}t^3 + t \\ \frac{2}{3}\left(2t^2 + \frac{1}{4}\right)^{3/2} \end{bmatrix}, \qquad 0 \le t \le 2.$$

Solution:

Step 1. Compute $x'(t)$ and $y'(t)$:

$$x'(t) = 7t^2 + 1 \quad \text{and} \quad y'(t) = \left(2t^2 + \frac{1}{4}\right)^{\frac{1}{2}} \cdot 4t.$$

Step 2. Compute $[x'(t)]^2 + [y'(t)]^2$ and simplify.

$$\begin{aligned} [x'(t)]^2 + [y'(t)]^2 &= \left[7t^2 + 1\right]^2 + \left[\left(2t^2 + \frac{1}{4}\right)^{\frac{1}{2}} \cdot 4t\right]^2 \\ &= 49t^4 + 14t^2 + 1 + \left(2t^2 + \frac{1}{4}\right) \cdot 16t^2 \\ &= 49t^4 + 14t^2 + 1 + 32t^4 + 4t^2 \\ &= 81t^4 + 18t^2 + 1. \end{aligned}$$

Step 3. Recognize that you've got a perfect square:

$$81t^4 + 18t^2 + 1 = \left(9t^2 + 1\right)^2$$

(Step 3 doesn't always happen, but it's really nice when it does!)
Step 4. Plug what you get back into the formula from Theorem 18:

$$\begin{aligned}
\text{Arc Length} &= \int_0^2 \sqrt{\left[x'(t)\right]^2 + \left[y'(t)\right]^2} dt \\
&= \int_0^2 \sqrt{(9t^2 + 1)^2} dt \\
&= \int_0^2 (9t^2 + 1) dt \\
&= \left[3t^3 + t\right]_0^2 \\
&= 26.
\end{aligned}$$

We conclude that the arc length is 26.

End of Example 21. $\square$

6 2026-09-04 | Week 02 | Lecture 06

Reading assignment: Chapter 11.3

6.1 Smoothness of parameterized curves

Definition 22 (C^1 function)**.** A function f is said to be ***continuously differentiable*** on the interval $[a, b]$ if f is differentiable on $[a, b]$ and f' is continuous. As shorthand, we write

$$f \in C^1[a, b].$$

Definition 23 (Smooth)**.** A curve C parameterized by

$$\mathbf{r}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}, \qquad a \le t \le b$$

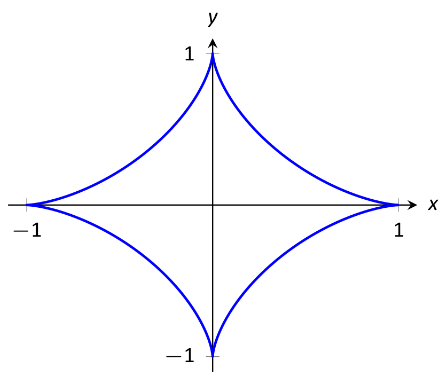
is ***smooth*** if $x, y \in C^1([a, b])$ and $x'(t)$ and $y'(t)$ are never simultaneously zero.

A curve is ***piecewise smooth*** on $[a, b]$ if it can be partitioned into subintervals such that C is smooth on each subinterval.

Example 24 (A non-smooth curve)**.** Consider the equations

$$\begin{cases} x(t) := \cos^3(t) \\ y(t) := \sin^3(t) \end{cases} \qquad 0 \le t \le 2\pi.$$

This has graph



Astroid
$x = \cos^3 t$
$y = \sin^3 t$

This sure doesn't look smooth, because it has pointy cusps! Let's show that using the definition.

First, differentiate:

$$\begin{cases} x'(t) = -3\cos^2(t)\sin(t) \\ y'(t) = 3\sin^2(t)\cos(t) \end{cases}$$

Are these functions ever simultaneously zero? Yes, when t takes one of the following values:

$$0, \quad \frac{\pi}{2}, \quad \pi, \quad \frac{3\pi}{2}, \quad \text{or} \quad 2\pi.$$

Therefore the curve is not smooth (although it is piecewise smooth).

End of Example 24. □

6.2 Parameterized curves: tangent line and normal line

Let's change notation, and suppose the ant is walking along the curve C parameterized by

$$\mathbf{r}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}, \qquad a \le t \le b.$$

At any given instant the ant is facing some direction—the direction of $\mathbf{r}'(t)$—and we can draw a line in that direction. The derivative $\frac{dy}{dx}$ is the *slope of that line*.

How can we find $\frac{dy}{dx}$? The answer to this question is the following important idea:

$$\frac{dy}{dx}(t) = \frac{dy/dt}{dx/dt} = \frac{y'(t)}{x'(t)} \tag{8}$$

This idea so important I've put it in a box.

Equation (8) is valid provided that the following two assumptions hold:

- x and y are both differentiable (otherwise $x'(t)$ and $y'(t)$ wouldn't exist)
- $x'(t) \neq 0$ (otherwise we'd have divsion by zero)

Definition 25 (Tangent Line and Normal Line)**.** The ***tangent line*** to C at time $t = t_0$ is the line

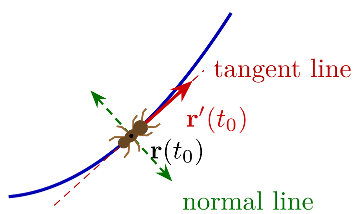
$$y - y(t_0) = \frac{y'(t_0)}{x'(t_0)} \left(x - x(t_0)\right),$$

provided that $x'(t_0) \neq 0$.

The ***normal line*** to C at time $t = t_0$ is the line

$$y - y(t_0) = -\frac{x'(t_0)}{y'(t_0)} \left(x - x(t_0)\right),$$

provided that $y'(t_0) \neq 0$.



Remark 26. *Both the tangent line and the normal line pass through the point $(x, y) = (x(t), y(t))$, which lies on the curve. The only difference is their slope. The tangent line has slope*

$$m = \frac{dy}{dx}(t_0)$$

while the normal line has slope

$$m_{\text{norm}} = -\frac{1}{m}.$$

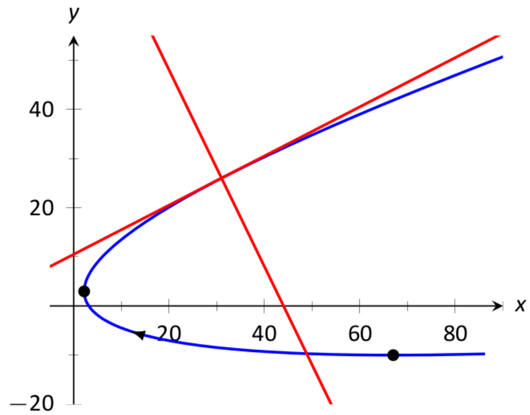
The two lines are perpendicular to each other: the tangent line lies tangent to the curve (parallel to the the velocity $\mathbf{r}'(t_0)$), while the normal line is perpendicular to it.

End of Remark 26. ☐

Example 27 (Tangent and normal lines)**.** Let C be the curve defined by

$$\begin{cases} x(t) = 5t^2 - 6t + 4 \\ y(t) = t^2 + 6t - 1 \end{cases} \tag{9}$$

Problem: Find the tangent line and normal line to C at $t = 3$.
Solution: Before we start, this is the picture:



Differentiate:

$$\begin{cases} x'(t) = 10t - t \\ y'(t) = 2t + 6 \end{cases}$$

Therefore

$$\frac{dy}{dx}(t) = \frac{2t+6}{10t-6}. \tag{10}$$

Important note: $\frac{dy}{dx}$ is a function of t. It is not a function of x!

At the point $t = 3$, we have:

- $(x, y) = (31, 26)$ (by Eq. (9))
- The slope is $\frac{dy}{dx}(3) = \frac{1}{2}$ (by Eq. (10))

Combining these two facts, and using Definition 25, we conclude that the tangent line is

$$y - 26 = \frac{1}{2}(x - 31)$$

(this is the equation of a line in point-slope form). Here, we have $m_{\tan} = \frac{1}{2}$. The slope of the normal line is always $-\frac{1}{m_{\tan}}$, which is -2. We conclude that the normal line is

$$y - 26 = -2(x - 31).$$

End of Example 27. □